

LAB EXPERIMENT RECORD

Set 8: Employee Performance Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Saturday

AIM

To analyze employee performance data using **R programming** by creating a line chart, bar chart, and table, and to develop a **Tableau dashboard** for interactive exploration of employee performance.

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85



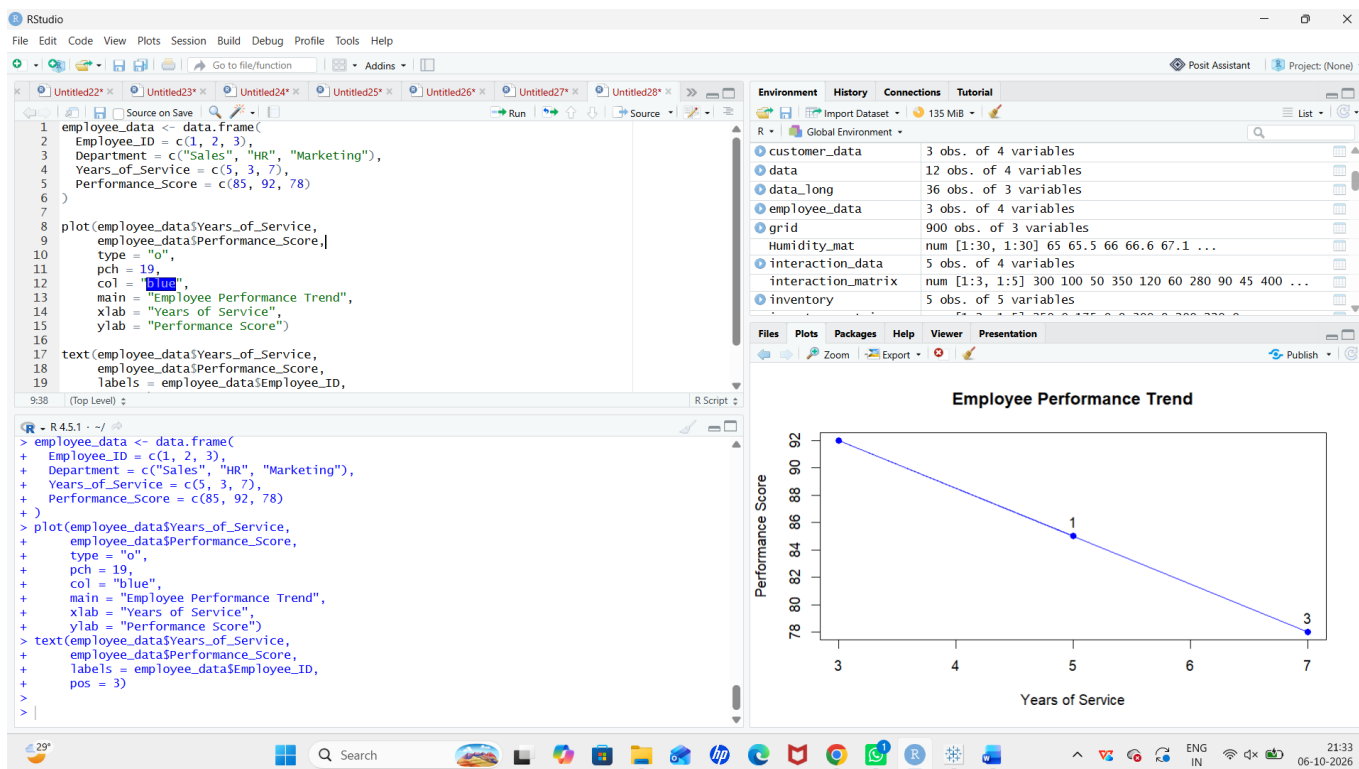
Employee ID	Department	Years of Service	Performance Score
2	HR	3	92
3	Marketing	7	78

1. Create a line chart to visualize the performance trend of employees over time.

```
employee_data <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)
```

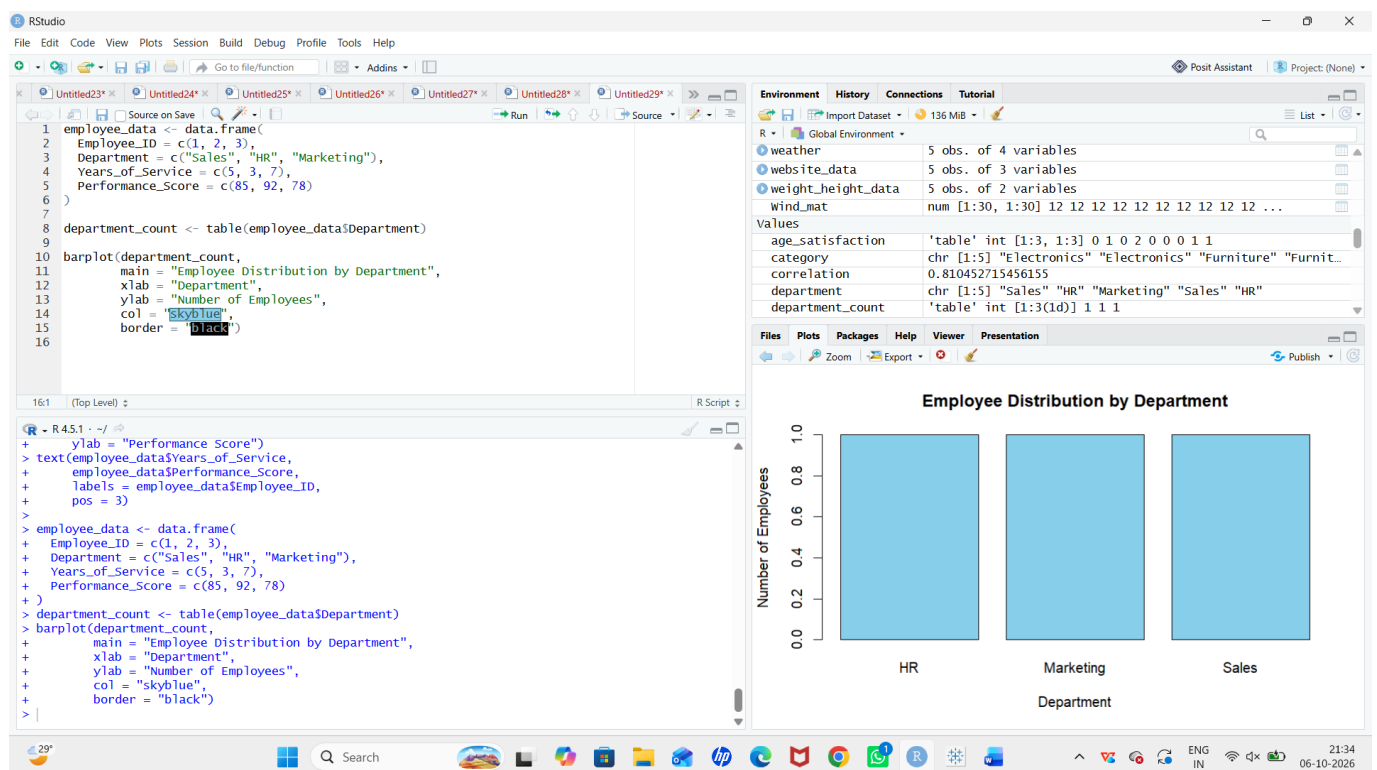
```
plot(employee_data$Years_of_Service,  
  employee_data$Performance_Score,  
  type = "o",  
  pch = 19,  
  col = "blue",  
  main = "Employee Performance Trend",  
  xlab = "Years of Service",  
  ylab = "Performance Score")
```

```
text(employee_data$Years_of_Service,  
  employee_data$Performance_Score,  
  labels = employee_data$Employee_ID,  
  pos = 3)
```



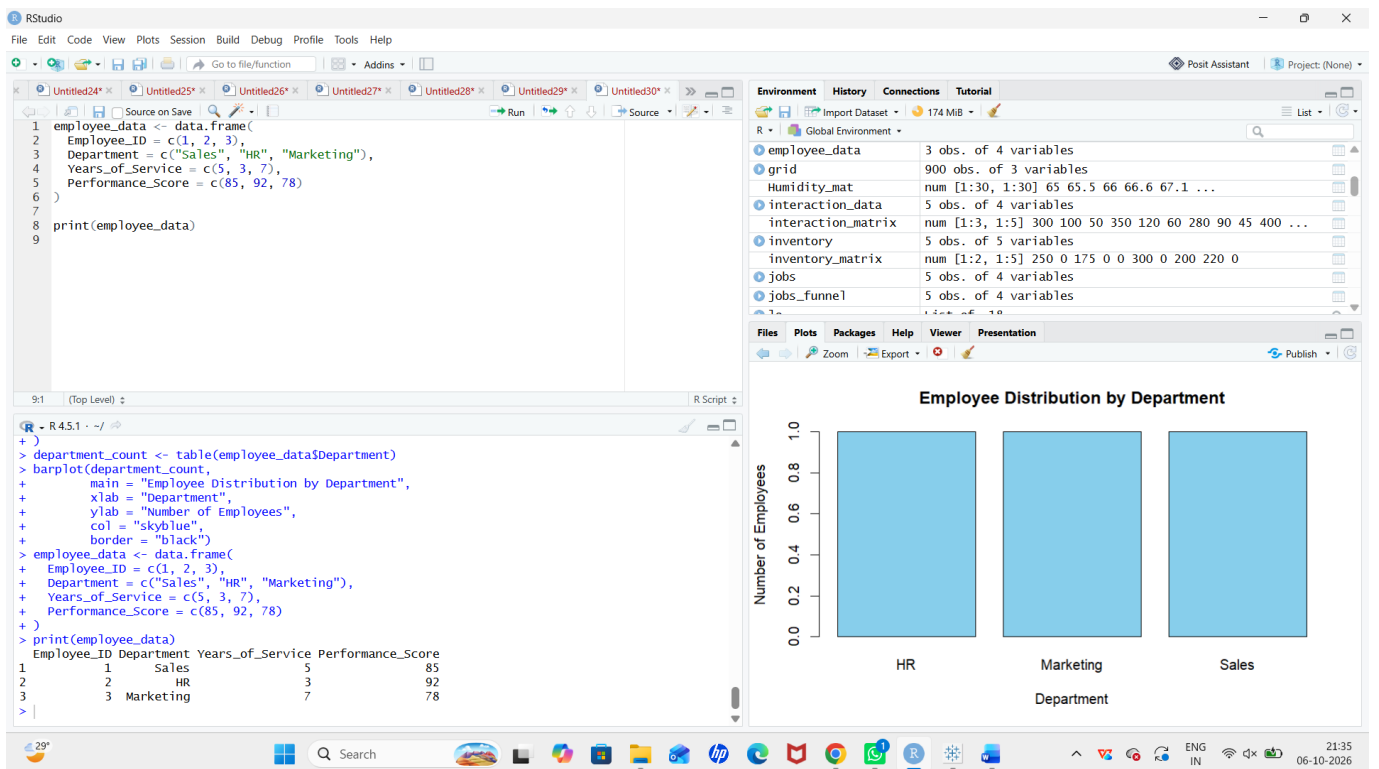
2. Generate a bar chart showing the distribution of employees across different departments.

```
employee_data <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
  
department_count <- table(employee_data$Department)  
  
barplot(department_count,  
  main = "Employee Distribution by Department",  
  xlab = "Department",  
  ylab = "Number of Employees",  
  col = "skyblue",  
  border = "black")
```

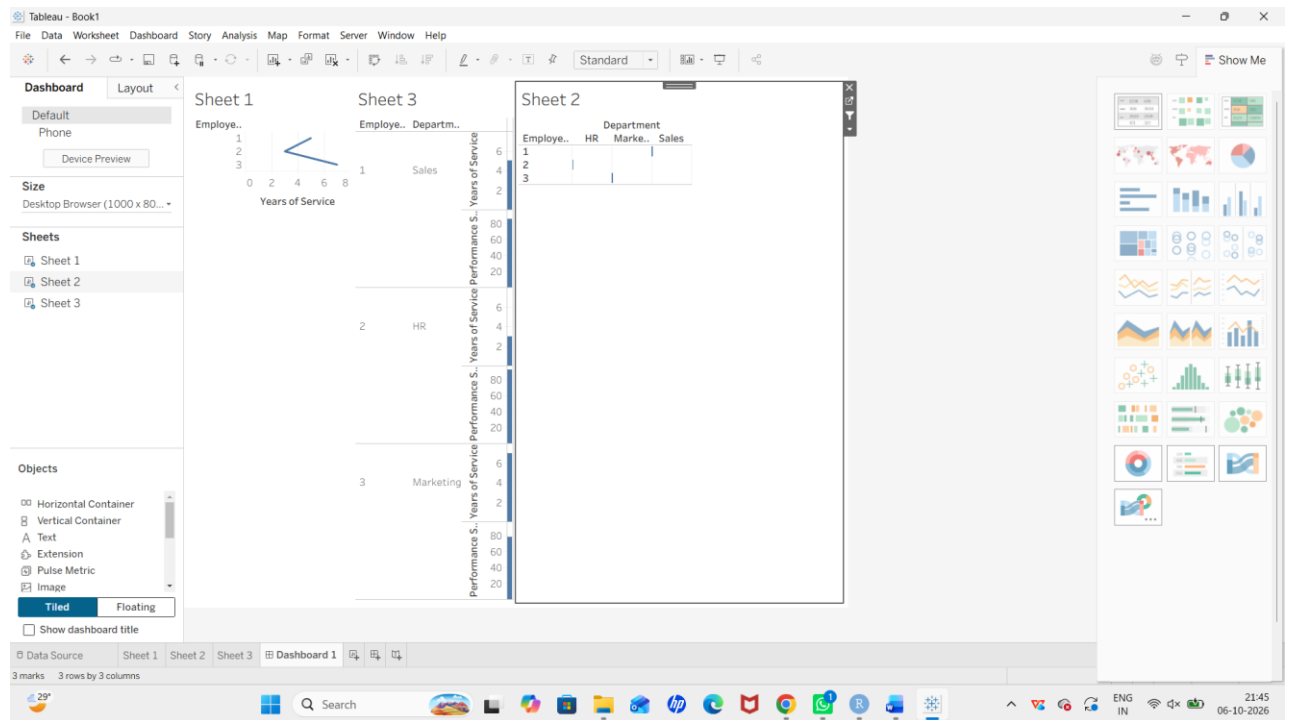


3. Build a table to display the performance data for each employee. Label the table elements.

```
employee_data <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
  
print(employee_data)
```



14.



PROCEDURE

1. Create the Employee Performance dataset in R with Employee ID, Department, Years of Service, and Performance Score.
2. Create a line chart to visualize the performance score of employees based on their years of service.
3. Add suitable axis labels and title to the line chart.
4. Count the number of employees in each department.
5. Create a bar chart to show the distribution of employees across Sales, HR, and Marketing departments.
6. Add appropriate labels and title to the bar chart.
7. Create a table displaying Employee ID, Department, Years of Service, and Performance Score.
8. Import the employee performance dataset into Tableau.
9. Create the line chart in Tableau using Years of Service and Performance Score.
10. Create the bar chart in Tableau to show the number of employees in each department.
11. Create a table to display the complete employee performance information.
12. Combine the line chart, bar chart, and table into a Tableau dashboard.
13. Add filters to make the dashboard interactive.

RESULT

The employee performance data was successfully analyzed using R programming. The line chart showed the performance trend based on years of service, while the bar chart displayed the number of employees in each department. A table was created to display the complete employee performance data. Finally, a Tableau dashboard was developed by combining the line chart, bar chart, and table for interactive analysis.

